

RM10

## Numeracy II: Mensuration, Averages, Statistics, Data Interpretation, Probability

*Numeracy II supplies roughly 8 to 10 of the 19-21 reasoning questions - mensuration and data interpretation are the two largest sub-blocks, with probability (dice especially) and calendar-clock giving one each almost every year. Pie-chart DI appeared in RAS Pre 2013, 2018 and 2024; dice probability in RAS Pre 2021, 2023 and 2024. Target full marks here in about ten minutes.*

Reasoning and Mental Ability gives you 19 to 21 questions in every RAS Prelims paper, and it is the easiest block in the whole exam to convert at 90 per cent - because nothing here has to be remembered, only practised. This chapter alone, covering mensuration, averages, statistics, data interpretation, probability, speed, work, calendar and clock, is worth more marks than most of the Rajasthan content chapters. Notes for it teach METHOD, not facts.

## Numeracy II

### Mensuration 2D

- Square, rectangle, triangle
- Heron's formula
- Parallelogram, rhombus, trapezium
- Circle, semicircle, sector
- Path around a field

### Mensuration 3D

- Cube and cuboid
- Cylinder
- Cone (slant height)
- Sphere and hemisphere
- Change of dimension:  $k$ ,  $k$  squared,  $k$  cubed

### Averages

- Sum divided by count
- Removal, inclusion, replacement
- Weighted average
- Average speed  $2xy/(x+y)$

### Statistics

- Mean,  $\Sigma fx / \Sigma f$
- Median from cumulative frequency
- Mode
- Range
- Mode =  $3 \text{ Median} - 2 \text{ Mean}$

### Data interpretation

- Text tables
- Bar charts
- Pie charts:  $1 \text{ degree} = \text{total}/360$
- Average, ratio, percentage change

### Counting and chance

- $nPr$  order matters
- $nCr$  order does not
- Repeated letters  $n!/(p!q!)$
- Coins, dice, cards, balls

### Speed and work

- $5/18$  and  $18/5$
- Trains: pole vs platform
- Relative speed
- One day's work  $1/x$
- Pipes and leaks

### Calendar and clock

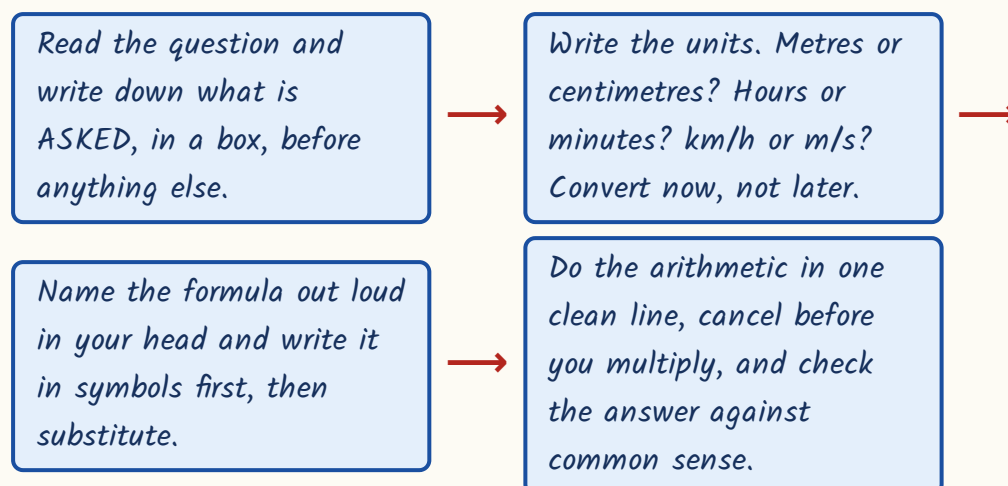
- Odd days
- 400 years = 0 odd days
- Month codes
- Angle =  $|30H - 5.5M|$

## How to attack Numeracy II - read this before the formulas

Reasoning and Mental Ability gives 19-21 questions in every RAS Prelims paper. This one chapter - mensuration, averages, statistics, data interpretation, probability, speed, work, calendar and clock - carries more marks than most of the Rajasthan content chapters you sweat over. And unlike them, nothing here has to be remembered from a book. Every answer is manufactured on the spot from a formula plus twenty seconds of arithmetic. That is why a trained candidate converts this block at 90 per cent while an untrained one loses eight marks to silly slips.

- Mensuration and DI are the two biggest sub-blocks - do them first in the paper.
- Probability with dice repeats most often; RAS Pre 2021, 2023 and 2024 all carried a dice question.
- Pie chart DI appeared in RAS Pre 2013, 2018 and 2024 - the data is always given in the stem.
- Calendar and clock give one question that takes 40 seconds if you know the two formulas.
- Nothing in this chapter needs a calculator. If your working needs one, your method is wrong.

### The four-step attack on any numeracy question



- Cancel first, multiply later -  $22/7$  with a radius of 7, 14 or 21 always cancels.
- Use  $\pi = 22/7$  whenever the radius is a multiple of 7; otherwise 3.14 is intended.
- Never average two averages, never average two speeds, never subtract two times - three of the most common traps in this chapter.
- If an option equals your answer times 2, or divided by 2, suspect a missing one-half in your formula.
- Sanity check every answer: a joint work time must be less than either worker alone; an average must lie between the smallest and largest value.

## Plane figures - perimeter and area

The commonest mensuration question gives you one measurement of a flat figure and asks for another. Perimeter of a square, so what is the area? Three sides of a triangle, so what is the area? Diameter of a circle, so what is the area? There is no trick - only the formula and one clean substitution. The marks are lost at the substitution step, not at the recall step, so slow down there.

### Plane figures - the full formula sheet

| Figure                                | Perimeter / circumference          | Area                                                                   |
|---------------------------------------|------------------------------------|------------------------------------------------------------------------|
| Square, side $a$                      | $4a$ (diagonal $= a\sqrt{2}$ )     | $a^2$ , or $d^2/2$ from the diagonal                                   |
| Rectangle, $l \times b$               | $2(l + b)$                         | $lb$ (diagonal $= \sqrt{l^2 + b^2}$ )                                  |
| Triangle, base $b$ , height $h$       | $a + b + c$                        | $\frac{1}{2} \times \text{base} \times \text{height}$                  |
| Triangle, three sides known           | $a + b + c$                        | Heron: $\sqrt{[s(s-a)(s-b)(s-c)]}$ , $s = (a+b+c)/2$                   |
| Equilateral triangle, side $a$        | $3a$                               | $(\sqrt{3}/4)a^2$ , height $= (\sqrt{3}/2)a$                           |
| Parallelogram, base $b$ , height $h$  | $2(\text{sum of adjacent sides})$  | base $\times$ height - NO one-half                                     |
| Rhombus, diagonals $d_1, d_2$         | $4 \times \text{side}$             | $\frac{1}{2} \times d_1 \times d_2$                                    |
| Trapezium, parallel sides $a$ and $b$ | sum of all four sides              | $\frac{1}{2} \times (a + b) \times \text{distance between them}$       |
| Circle, radius $r$                    | $2\pi r$                           | $\pi r^2$                                                              |
| Semicircle, radius $r$                | $\pi r + 2r$                       | $\frac{1}{2}\pi r^2$                                                   |
| Sector, angle $\theta$                | arc $= (\theta/360) \times 2\pi r$ | $(\theta/360) \times \pi r^2 = \frac{1}{2} \times \text{arc} \times r$ |

Worked example - Heron's formula. The sides of a triangle are 13 cm, 14 cm and 15 cm. Find the area. You are given three sides and no height, so Heron is the only route. Learn this 13-14-15 triangle by heart; RPSC and every other exam reuse it because the answer comes out as a whole number.

- $s = (a + b + c)/2 = (13 + 14 + 15)/2 = 42/2 = 21$ .
- $s - a = 21 - 13 = 8$ ;  $s - b = 21 - 14 = 7$ ;  $s - c = 21 - 15 = 6$ .
- $\text{Area} = \sqrt{[s(s-a)(s-b)(s-c)]} = \sqrt{(21 \times 8 \times 7 \times 6)}$ .
- Group the numbers before multiplying:  $21 \times 6 = 126$  and  $8 \times 7 = 56$ , so the product is  $126 \times 56 = 7056$ .
- $\sqrt{7056} = 84$ . Area =  $84 \text{ cm}^2$ .
- Second example, same section: sector of radius 21 cm, angle 120 degrees,  $\pi = 22/7$ . Area =  $(120/360) \times (22/7) \times 21 \times 21 = (1/3) \times 1386 = 462 \text{ cm}^2$ . The full circle would be  $1386 \text{ cm}^2$  and the ARC alone is only 44 cm - three different answers sitting in the options.

#### TRAP

The missing one-half. Triangle, trapezium and rhombus all carry a one-half; parallelogram does NOT. Trapezium with parallel sides 22 and 14 and distance 9: area =  $\frac{1}{2} \times (22 + 14) \times 9 = 162 \text{ cm}^2$ . Dropping the half gives 324 and using the difference  $(22 - 14)$  gives 36 - both will be sitting in the options waiting for you.

**Paths around a field, and what happens when you change a dimension**

Two examiner favourites live here. First, a path of uniform width laid around or inside a rectangular field - you must find the area of the path itself, not of the field. Second, a percentage change question: the side goes up by 20 per cent, by how much does the area go up? Both are one-line problems once you stop drawing and start substituting.

### Paths and dimension-change rules

| Situation                                          | Formula                                                     | Note                                              |
|----------------------------------------------------|-------------------------------------------------------------|---------------------------------------------------|
| Path of width $w$ OUTSIDE a rectangle $l \times b$ | $2w(l + b) + 4w^2$                                          | the $+4w^2$ is the four corner squares            |
| Path of width $w$ INSIDE a rectangle $l \times b$  | $2w(l + b) - 4w^2$                                          | corners are subtracted here                       |
| Two roads of width $w$ crossing at right angles    | $w(l + b) - w^2$                                            | the $-w^2$ stops the crossing being counted twice |
| Every linear dimension multiplied by $k$           | length $\times k$ , area $\times k^2$ , volume $\times k^3$ | the single most useful rule in mensuration        |
| Two dimensions change by $x\%$ and $y\%$           | net change $= x + y + xy/100$                               | use a minus sign for a decrease                   |
| Three dimensions change by $x\%$ each              | apply the two-dimension rule twice                          | or simply cube the multiplier                     |

Worked example - path outside a field. A rectangular field is 50 m long and 40 m wide. A path 3 m wide is laid all around it on the OUTSIDE. Find the area of the path. Do it both ways once, so that you trust the shortcut afterwards.

- Long way. Outer rectangle =  $(50 + 2 \times 3)$  by  $(40 + 2 \times 3) = 56 \times 46 = 2576 \text{ m}^2$ .
- Inner field =  $50 \times 40 = 2000 \text{ m}^2$ .
- Path =  $2576 - 2000 = 576 \text{ m}^2$ .
- Short way.  $2w(l + b) + 4w^2 = 2 \times 3 \times (50 + 40) + 4 \times 9 = 540 + 36 = 576 \text{ m}^2$ .
- Note the width is added TWICE to each dimension - once on each side. Adding it once gives  $53 \times 43$  and a wrong answer.
- Forgetting the four corner squares gives  $540 \text{ m}^2$ , the classic distractor.
- Square side up 20%: area change =  $20 + 20 + (20 \times 20)/100 = 44\%$  increase. Check: 100 to 120 means 10000 to 14400.
- Radius up 10%: area up 21%. Radius down 10%: area down 19%, not 20%.
- Side up 20% and down 20% in turn: net -4%, never zero.
- Edge of a cube up 50%: multiplier 1.5, so surface area  $\times 2.25$  (up 125%) and volume  $\times 3.375$  (up 237.5%).
- Radius of a sphere doubled: surface area  $\times 4$ , volume  $\times 8$ . Halved: surface  $\times 1/4$ , volume  $\times 1/8$ .

## Solids - surface area and volume

Six solids, and RPSC has asked all of them. The question either gives a dimension and asks for volume or surface area, or gives the volume and works backwards to a dimension. Everything you need sits in one table. The only genuine piece of thinking is the cone, where the slant height  $l$  is almost never given - you compute it from  $r$  and  $h$  by Pythagoras.

### Solids - volume, curved surface, total surface

| Solid          | Volume | Curved / lateral SA | Total SA                         |
|----------------|--------|---------------------|----------------------------------|
| Cube, edge $a$ | $a^3$  | $4a^2$              | $6a^2$ (diagonal = $a\sqrt{3}$ ) |

| Solid                             | Volume           | Curved / lateral SA                     | Total SA                        |
|-----------------------------------|------------------|-----------------------------------------|---------------------------------|
| Cuboid $l \times b \times h$      | $lbh$            | $2h(l + b)$                             | $2(lb + bh + hl)$               |
| Cylinder, radius $r$ , height $h$ | $\pi r^2 h$      | $2\pi rh$                               | $2\pi r(r + h)$                 |
| Cone, radius $r$ , height $h$     | $(1/3)\pi r^2 h$ | $\pi rl$ , where $l = \sqrt{r^2 + h^2}$ | $\pi r(l + r)$                  |
| Sphere, radius $r$                | $(4/3)\pi r^3$   | —                                       | $4\pi r^2$                      |
| Hemisphere (solid), radius $r$    | $(2/3)\pi r^3$   | $2\pi r^2$                              | $3\pi r^2$ (curved + flat face) |

Worked example - the cone. A cone has radius 7 cm and vertical height 24 cm. Find its curved surface area, taking  $\pi = 22/7$ . Notice what is given: the VERTICAL height, not the slant height. The formula  $\pi rl$  needs  $l$ , so the first line of your solution must be Pythagoras.

- $l = \sqrt{r^2 + h^2} = \sqrt{7^2 + 24^2} = \sqrt{49 + 576} = \sqrt{625} = 25 \text{ cm.}$
  - 7-24-25 is a Pythagorean triple - memorise it along with 3-4-5, 5-12-13, 8-15-17, 9-40-41.
  - Curved surface area =  $\pi rl = (22/7) \times 7 \times 25$ . The 7 cancels the denominator straight away.
  - $= 22 \times 25 = 550 \text{ cm}^2$ .
  - Total surface area, if that had been asked, =  $\pi r(l + r) = (22/7) \times 7 \times 32 = 704 \text{ cm}^2$ .
  - Volume, if that had been asked, =  $(1/3) \times (22/7) \times 49 \times 24 = 1232 \text{ cm}^3$ .
- Three different numbers - read the question word for word.



### The cone - one shape, four different answers

#### Radius $r$

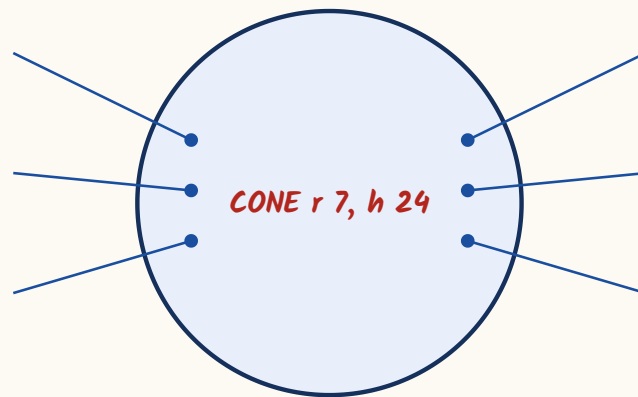
7 cm - the base circle

#### Slant height $l$

$\sqrt{49 + 576} = 25$  cm

#### Total SA

$\pi r (l + r) = 704$  cm<sup>2</sup>



#### Vertical height $h$

24 cm - given, not the slant

#### Curved SA

$\pi r l = 550$  cm<sup>2</sup>

#### Volume

$(1/3) \pi r^2 h = 1232$  cm<sup>3</sup>

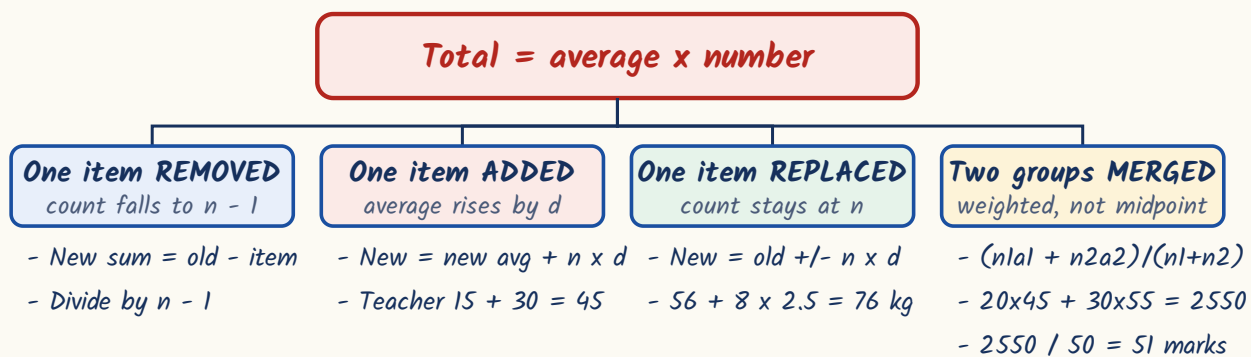
### TRAP

Curved versus total versus volume. Three of the four options in a solids question are the OTHER two quantities for the same solid. A solid hemisphere of radius 7 has curved SA 308, total SA 462 and the full sphere 616 - all three will be offered. Also watch the units: cm<sup>2</sup> for any area, cm<sup>3</sup> for any volume. If the option carries the wrong unit, it is wrong however good the number looks.

### Averages - including weighted average and average speed

Average is just total divided by count, but RPSC never asks it that plainly. The real questions are of four shapes: an item is removed, an item is added, an item is replaced, or two groups of different sizes are merged. In every one of them the trick is the same - stop thinking about averages and go back to the TOTAL. Total = average x number of items. Change the total, then divide again.

### Four shapes of an average question



### Average - the rules that answer the standard shapes

| Situation                                 | Rule                                               | Worked value                  |
|-------------------------------------------|----------------------------------------------------|-------------------------------|
| Basic                                     | Average = sum $\div$ number of items               | —                             |
| First $n$ natural numbers                 | $(n + 1)/2$                                        | first 10 give 5.5             |
| First $n$ odd numbers                     | $n$                                                | first 10 odd numbers give 10  |
| First $n$ even numbers                    | $n + 1$                                            | first 10 even numbers give 11 |
| One item REMOVED                          | new sum = old sum - item, then $\div (n-1)$        | —                             |
| One item ADDED, average rises by $d$      | new item = new average + $n \times d$              | —                             |
| One item REPLACED, average changes by $d$ | new item = old item $\pm n \times d$               | —                             |
| Two groups merged                         | weighted average = $(n_1a_1 + n_2a_2)/(n_1 + n_2)$ | —                             |
| Equal DISTANCES at speeds $x$ and $y$     | $2xy/(x + y)$ , the harmonic mean                  | 60 and 40 give 48             |
| Equal TIMES at speeds $x$ and $y$         | $(x + y)/2$ , the arithmetic mean                  | 60 and 40 give 50             |

**Worked example - weighted average.** In a class of 50 students, the 20 boys score an average of 45 marks and the 30 girls an average of 55 marks. Find the average marks of the whole class. The tempting answer is the midpoint of 45 and 55, which is 50. It is wrong, and it is option (a) in every version of this question.

- Total marks of the boys =  $20 \times 45 = 900$ .
- Total marks of the girls =  $30 \times 55 = 1650$ .
- Total marks of the class =  $900 + 1650 = 2550$ .
- Class average =  $2550 \div 50 = 51$  marks.
- Why 51 and not 50: there are more girls, and the girls score higher, so the combined average is pulled towards the girls' 55.
- Sanity check that costs one second - the answer must lie between 45 and 55, and closer to the bigger group's value.

### Average speed - equal distance versus equal time

| Equal DISTANCE each way                         | Equal TIME at each speed                    |
|-------------------------------------------------|---------------------------------------------|
| 'goes to B and returns', 'up and down the hill' | 'drove for 2 hours at x, then 2 hours at y' |
| Use $2xy/(x + y)$ - the harmonic mean           | Use $(x + y)/2$ - the plain arithmetic mean |
| 60 and 40 give 48 km/h                          | 60 and 40 give 50 km/h                      |
| Answer is always BELOW the simple midpoint      | Answer is exactly the midpoint              |
| More time is spent at the slower speed          | Time is shared equally by definition        |

**YAAD RAKHO**

Remember - the replacement rule. When one person swaps for another in a group of  $n$  and the average moves by  $d$ , the whole change  $n \times d$  is carried by that single swap. Eight people, average up 2.5 kg, the person leaving weighs 56 kg: the new person =  $56 + (8 \times 2.5) = 76$  kg. Adding 2.5 only once gives 58.5 - the option that catches half the hall. Same logic for inclusion: 30 students average 14, teacher raises it by 1, so teacher =  $15 + 30 = 45$  years.

## Mean, median, mode and range

Three averages of central tendency plus one measure of spread. Mean is the arithmetic average, median is the middle value after sorting, mode is the value that occurs most often, range is largest minus smallest. RPSC asks these from an ungrouped list of six or seven numbers, or from a small frequency table. The frequency-table version is where the marks are lost, because students average the values and ignore the frequencies.

### Central tendency and dispersion

| Measure            | Ungrouped data                                    | Grouped / frequency data                                         |
|--------------------|---------------------------------------------------|------------------------------------------------------------------|
| Mean               | $\text{sum} \div n$                               | $\Sigma fx \div \Sigma f$ - multiply each value by its frequency |
| Median, $n$ odd    | the $((n+1)/2)$ th value after SORTING            | the value at the cumulative frequency crossing $N/2$             |
| Median, $n$ even   | mean of the $(n/2)$ th and $(n/2 + 1)$ th values  | average of the two middle observations                           |
| Mode               | the most frequently occurring value               | the value with the highest frequency                             |
| Empirical relation | $\text{Mode} = 3 \text{ Median} - 2 \text{ Mean}$ | for a moderately skewed distribution                             |

| Measure | Ungrouped data    | Grouped / frequency data                 |
|---------|-------------------|------------------------------------------|
| Range   | maximum – minimum | a measure of dispersion, not of position |

Worked example - mean of a frequency distribution. Value x is 10, 20, 30, 40, 50 with frequencies 3, 5, 8, 3, 1 respectively. Find the arithmetic mean. Make two columns on your rough sheet - one for f, one for fx - and total both.

- $\Sigma f = 3 + 5 + 8 + 3 + 1 = 20$  observations in all.
- fx values:  $10 \times 3 = 30$ ,  $20 \times 5 = 100$ ,  $30 \times 8 = 240$ ,  $40 \times 3 = 120$ ,  $50 \times 1 = 50$ .
- $\Sigma fx = 30 + 100 + 240 + 120 + 50 = 540$ .
- $\text{Mean} = \Sigma fx \div \Sigma f = 540 \div 20 = 27$ .
- Averaging 10, 20, 30, 40, 50 straight gives 30 - wrong, because it pretends every value occurred equally often.

Worked example - median from a cumulative frequency. A table gives the number of children per family in 50 families: 0 children in 6 families, 1 child in 10, 2 children in 12, 3 children in 14, 4 children in 8. Find the MEDIAN number of children. The cumulative frequency column does all the work.

- Cumulative frequencies: 6, then 16, then 28, then 42, then 50.
- $N = 50$ , which is even, so the median is the average of the 25th and 26th observations.
- The 17th to 28th observations all lie in the '2 children' row, because the cumulative jumps from 16 to 28 there.
- So the 25th and 26th are both 2.  $\text{Median} = (2 + 2)/2 = 2$ .
- The MODE here is 3, because 14 is the highest frequency. Median 2, mode 3 - two different answers in the same table.
- Empirical relation check, on a different data set: mean 26 and median 28 give  $\text{mode} = 3(28) - 2(26) = 84 - 52 = 32$ .

## Data interpretation - tables, bar charts and pie charts

A DI question hands you a small data set and asks two or three arithmetic questions about it. In RAS the set is tiny - five rows at most - and it is written out in words in the stem, so you never have to read a graph badly. The arithmetic is only average, ratio, difference and percentage. What kills candidates is misreading which row the question wants. Worked example - a pie chart. The monthly expenditure of a family totalling ₹36,000 is shown by a pie chart with these central angles: Food 120 degrees, House rent 90 degrees, Education 60 degrees, Transport 45 degrees, Miscellaneous 45 degrees. Find (a) the spend on food, (b) by how much house rent exceeds education, and (c) house rent plus transport as a percentage of the total.

**The pie chart - 360 degrees stands for Rs 36,000**

**Food 120 deg**

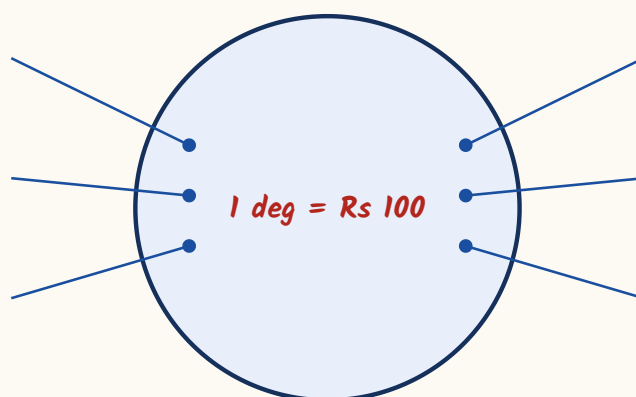
Rs 12,000 - one third

**Education 60 deg**

Rs 6,000

**Miscellaneous 45 deg**

Rs 4,500



**House rent 90 deg**

Rs 9,000 - one quarter

**Transport 45 deg**

Rs 4,500

**Check the angles**

$120 + 90 + 60 + 45 + 45 = 360$

- First line of every pie question: 360 degrees stands for the whole, here ₹36,000. So 1 degree =  $36000 \div 360 = ₹100$ .
- (a) Food =  $120 \times 100 = ₹12,000$ . Equivalently  $120/360 =$  one third of ₹36,000.
- (b) House rent =  $90 \times 100 = ₹9,000$ ; education =  $60 \times 100 = ₹6,000$ ; difference = ₹3,000. Faster:  $(90 - 60) \times 100 = ₹3,000$ .
- (c) House rent + transport =  $90 + 45 = 135$  degrees. Percentage =  $(135/360) \times 100 = 37.5\%$ .
- For a percentage question you never need the rupees at all - work in degrees and divide by 360.
- Check that the angles add to 360:  $120 + 90 + 60 + 45 + 45 = 360$ . If they do not, you have misread a row.

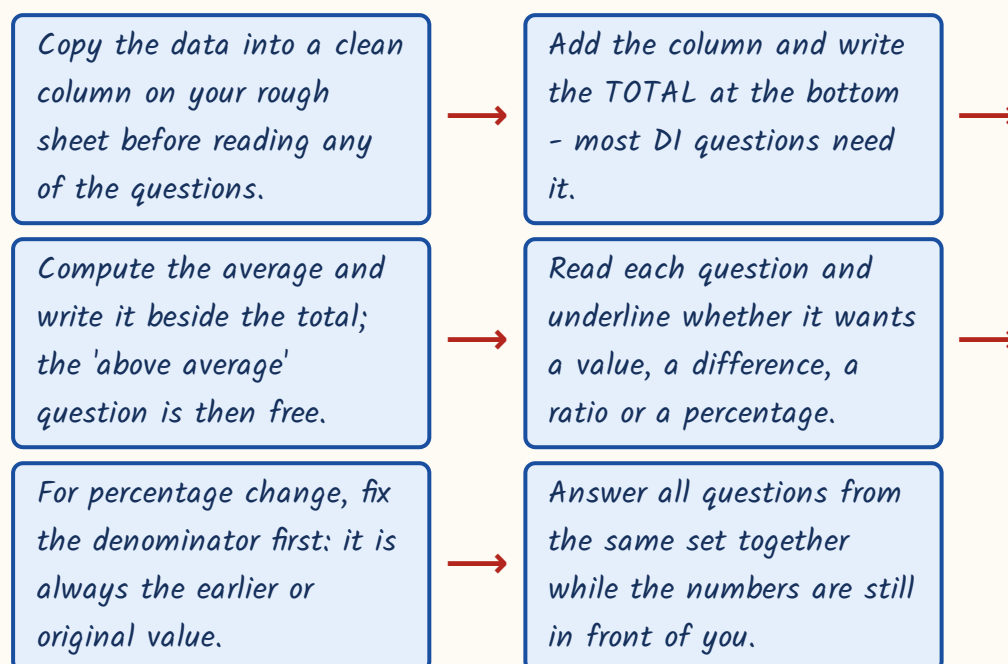
#### ASKED IN RAS

Pie-chart DI on household or budget expenditure appeared in RAS Pre 2013, 2018 and 2024. The wording changes, the method never does: convert 360 degrees to the given total, get the value of one degree, then answer. Expect two questions from the same chart.

Worked example - a table. The wheat production of a district in lakh tonnes was 24 in 2019, 30 in 2020, 27 in 2021, 36 in 2022 and 33 in 2023. Three questions get asked from a table like this: the average, a percentage change between two years, and a count of years above or below the average. A bar chart is the same data drawn as five bars, and it asks the same three questions - never redraw it, just read the heights into a column. A second, double-bar type gives boys and girls in five schools - A 240 and 160, B 180 and 220, C 300 and 200, D 150 and 150, E 210 and 190 - and asks for a ratio or for which school has the highest percentage of girls.

- Average production =  $(24 + 30 + 27 + 36 + 33) \div 5 = 150 \div 5 = 30$  lakh tonnes.
- Percentage rise 2019 to 2020 =  $(30 - 24)/24 \times 100 = 6/24 \times 100 = 25\%$ .  
Always divide by the ORIGINAL year, not the new one - dividing by 30 gives the wrong 20%.
- Years ABOVE the average of 30: 24 no, 30 is equal and therefore not above, 27 no, 36 yes, 33 yes. Answer: 2 years.
- Schools table - total boys =  $240+180+300+150+210 = 1080$ ; total girls =  $160+220+200+150+190 = 920$ ; ratio  $1080 : 920 = 27 : 23$  after dividing by 40.
- Highest percentage of girls: A  $160/400 = 40\%$ , B  $220/400 = 55\%$ , C  $200/500 = 40\%$ , D  $150/300 = 50\%$ , E  $190/400 = 47.5\%$ . Answer: School B.
- Note that School C has 200 girls, more than D's 150, yet a lower percentage. Largest NUMBER is not largest SHARE - that is the whole point of the question.

### How to work a DI set in under two minutes



## Permutation and combination



One question, sometimes two, and it is always decided in the first three seconds by a single decision: does the ORDER matter? Arrangements, seatings, words made from letters, numbers made from digits - order matters, use permutation. Selections, committees, teams, handshakes, chords, groups - order does not matter, use combination. Get that decision right and the arithmetic is trivial.

### Permutation and combination - formulas and standard results

| Thing asked                            | Formula                                       | Standard value            |
|----------------------------------------|-----------------------------------------------|---------------------------|
| Arrange $r$ out of $n$ , order matters | $nPr = n!/(n-r)!$                             | $5P2 = 20$                |
| Select $r$ out of $n$ , order does not | $nCr = n!/[r!(n-r)!]$                         | $5C2 = 10$                |
| Arrange all $n$ distinct letters       | $n!$                                          | $DELHI = 5! = 120$        |
| Arrange $n$ letters with repeats       | $n!/(p! q! \dots)$                            | $BANANA = 6!/(3!2!) = 60$ |
| Handshakes / lines / chords among $n$  | $nC2 = n(n-1)/2$                              | 12 people give 66         |
| Diagonals of an $n$ -sided polygon     | $n(n-3)/2$                                    | octagon gives 20          |
| Useful identities                      | $nC0 = nCn = 1$ ; $nC1 = n$ ; $nCr = nC(n-r)$ | $10C7 = 10C3 = 120$       |
| Two independent choices                | MULTIPLY the counts                           | never add them            |

Worked example one - BANANA. In how many different ways can the letters of the word BANANA be arranged? Worked example two - a committee of 3 men and 2 women is to be formed from 6 men and 5 women; in how many ways? Do both, because between them they cover the two things that go wrong: repeated letters, and forgetting to multiply.

- BANANA has 6 letters: B once, A three times, N twice.
- Arrangements =  $6!/(3! \times 2!) = 720/(6 \times 2) = 720/12 = 60$ .
- Treating all six letters as distinct gives 720 - the standard wrong answer.
- Committee: choose the men first.  ${}^6C_3 = (6 \times 5 \times 4)/(3 \times 2 \times 1) = 120/6 = 20$  ways.
- Choose the women:  ${}^5C_2 = (5 \times 4)/(2 \times 1) = 20/2 = 10$  ways.
- The two choices are independent, so MULTIPLY:  $20 \times 10 = 200$  ways. Adding gives 30, which is the distractor.
- Handshake check on the same logic: 12 people, each pair shakes once, so  ${}^{12}C_2 = (12 \times 11)/2 = 66$ . Using  $12 \times 11 = 132$  counts every handshake from both sides.

### Permutation versus combination - decide in three seconds

| Permutation - ORDER matters                     | Combination - order does NOT matter        |
|-------------------------------------------------|--------------------------------------------|
| Arrange, seat, line up, form a word or a number | Select, choose, form a committee or a team |
| Prize positions: first, second, third           | Three winners with no ranking              |
| $nPr = n!/(n-r)!$                               | $nCr = n!/[r!(n-r)!]$                      |
| $nPr$ is always the LARGER of the two           | $nCr = nPr \div r!$                        |
| AB and BA are two different outcomes            | AB and BA are the same outcome             |
| Digits forming a number, letters forming a word | Handshakes, chords, lines through points   |

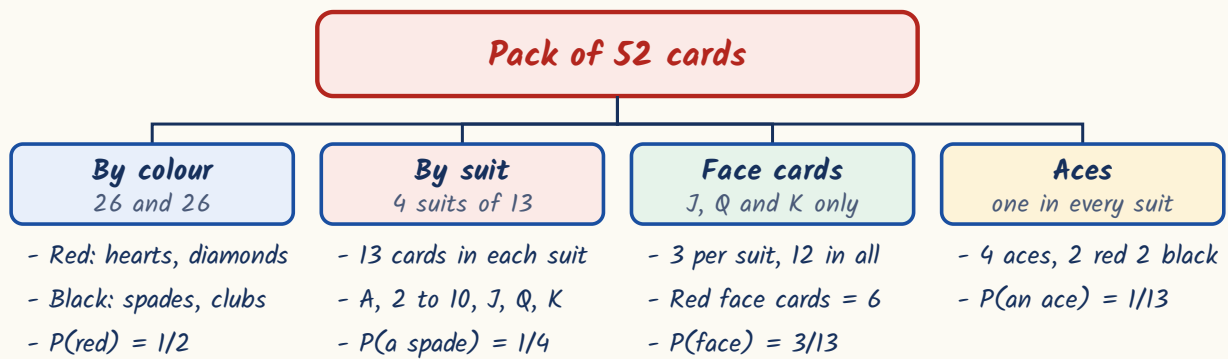
### Probability - coins, dice, cards and balls in a bag

Probability is favourable outcomes divided by total outcomes, and in RAS the total is always one of four small numbers:  $2$  to the power  $n$  for coins,  $36$  for two dice,  $52$  for cards, or the ball count in a bag. So the whole question reduces to counting the favourable cases correctly. Write the sample-space size FIRST, then count. Every answer must lie between  $0$  and  $1$  - if yours does not, you have inverted a fraction.

### Probability - the four standard settings

| Setting                              | Total outcomes                    | Things you must know                                                             |
|--------------------------------------|-----------------------------------|----------------------------------------------------------------------------------|
| $n$ coins tossed                     | $2^n$                             | exactly $r$ heads = $nCr \div 2^n$ ; 3 coins give 8 outcomes                     |
| One die                              | 6                                 | primes are 2, 3 and 5 - so $P(\text{prime}) = 1/2$ ; 1 is NOT prime              |
| Two dice                             | 36                                | doublets = 6, so $P = 1/6$ ; $P(\text{sum } 7) = 6/36 = 1/6$ , the likeliest sum |
| Two dice, ways to make a sum         | 2:1, 3:2, 4:3, 5:4, 6:5, 7:6      | 8:5, 9:4, 10:3, 11:2, 12:1 - it is a symmetric ladder                            |
| Pack of 52 cards                     | 52                                | 4 suits $\times$ 13; 26 red and 26 black; 4 aces                                 |
| Face cards                           | 12 in all                         | J, Q, K in each suit; red face cards = 6 only                                    |
| Balls drawn together, no replacement | $nC2$ for two balls               | $P(\text{both black from } 4W \text{ and } 5B) = 5C2 \div 9C2$                   |
| Balls drawn with replacement         | multiply the single probabilities | $(5/9)^2$ is NOT the same as $5C2 \div 9C2$                                      |
| Complement                           | $P(\text{not } E) = 1 - P(E)$     | use it for 'at least one'                                                        |

### A pack of 52 cards, cut three ways



Worked example one - two dice, sum of 8. Worked example two - a bag with 4 white and 5 black balls, two drawn together, both black. These two carry the two counting errors that RPSC exploits: forgetting that dice are ordered, and forgetting that 'drawn together' means without replacement.

- Two dice:  $n(S) = 6 \times 6 = 36$ , because the first die and the second die are distinguishable.
- Sum 8 comes from (2,6), (3,5), (4,4), (5,3), (6,2) - five ordered pairs.
- $P(\text{sum } 8) = 5/36$ . Counting the unordered sets 2+6, 3+5, 4+4 as only three cases gives the wrong  $1/12$ .
- Balls: total ways of drawing 2 from 9 =  ${}^9C_2 = (9 \times 8)/2 = 36$ .
- Ways of drawing 2 black from 5 =  ${}^5C_2 = (5 \times 4)/2 = 10$ .
- $P(\text{both black}) = 10/36 = 5/18$ . The distractor  $25/81$  comes from doing  $(5/9)^2$ , which would be right only WITH replacement.
- Third quick one: three coins, exactly two heads. Sample space 8; favourable HHT, HTH, THH;  $P = 3/8$ . Exactly three heads would be  $1/8$ .

**ASKED IN RAS**

A dice probability question appeared in RAS Pre 2021, 2023 and 2024 - three papers running. The forms used were  $P(\text{prime on one die}) = 1/2$ ,  $P(\text{doublet}) = 1/6$ ,  $P(\text{sum } 8) = 5/36$  and  $P(\text{sum a multiple of } 3) = 12/36 = 1/3$ . Learn the sum ladder and all four are instant.

**YAAD RAKHO**

Remember - the sum ladder for two dice reads 1, 2, 3, 4, 5, 6, 5, 4, 3, 2, 1 for sums 2 through 12, and adds to 36. So for any sum question, read the count straight off the ladder and divide by 36. Sum a multiple of 3 means sums 3, 6, 9, 12 - that is  $2 + 5 + 4 + 1 = 12$  ways, so  $1/3$ . For cards, remember 6 red face cards, not 12: the colour condition halves it.

## Time, speed and distance - including trains and boats

Speed = distance divided by time, and everything in this sub-topic is that one relation dressed differently. The two decisions that matter are the unit conversion and the choice of distance. A train crossing a pole covers only its own length; a train crossing a platform covers its own length PLUS the platform. Get those two right and the rest is division.

### Time, speed and distance - conversions and relative motion

| Situation      | Rule                                              | Example value                      |
|----------------|---------------------------------------------------|------------------------------------|
| Basic relation | $\text{Speed} = \text{Distance} \div \text{Time}$ | 210 km in 3 h gives 70 km/h        |
| km/h to m/s    | multiply by $5/18$                                | 54 km/h = 15 m/s; 72 km/h = 20 m/s |
| m/s to km/h    | multiply by $18/5$                                | 20 m/s = 72 km/h                   |

| Situation                           | Rule                                        | Example value                           |
|-------------------------------------|---------------------------------------------|-----------------------------------------|
| Train crossing a pole or a man      | distance = length of the train              | —                                       |
| Train crossing a platform or bridge | distance = train length + platform length   | —                                       |
| Two bodies moving OPPOSITE ways     | relative speed = sum of the speeds          | 5 and 3 give a gap of 8 km per hour     |
| Two bodies moving the SAME way      | relative speed = difference of the speeds   | 5 and 3 give a gap of 2 km per hour     |
| Boat downstream / upstream          | $u + v$ and $u - v$                         | $u$ = boat in still water, $v$ = stream |
| Same distance at two speeds         | distance is fixed, so time varies inversely | speeds 5 : 6 give times 6 : 5           |

Worked example one - a train 240 m long running at 72 km/h crosses a platform 360 m long; find the time. Worked example two - walking at 5 km/h a student reaches school 6 minutes late, walking at 6 km/h he is 4 minutes early; find the distance. The second is the harder standard type and it is worth learning as a pattern.

- Train: convert first.  $72 \text{ km/h} = 72 \times 5/18 = 20 \text{ m/s}$ .
- Distance covered = train + platform =  $240 + 360 = 600 \text{ m}$ .
- Time =  $600 \div 20 = 30$  seconds. Using only the 240 m gives 12 seconds, the trap answer.
- Late-and-early: the two walking times differ by  $6 + 4 = 10$  minutes =  $1/6$  hour.
- Let the distance be  $d$  km. Then  $d/5 - d/6 = 1/6$ .
- $(6d - 5d)/30 = 1/6$ , so  $d/30 = 1/6$ , giving  $d = 5 \text{ km}$ .
- Check: at 5 km/h it takes 60 minutes, at 6 km/h it takes 50 minutes - a gap of exactly 10 minutes.

- The late and the early minutes are ADDED to get the time gap, never subtracted.
- Two men walking apart at 5 and 3 km/h are  $8 \times 4 = 32$  km apart after 4 hours; walking the same way they would be only 8 km apart.
- Direct proportion shortcut: 210 km in 3 hours means 350 km takes  $3 \times (350/210) = 5$  hours, with no need to find the speed.
- A question in seconds and metres is a train question; a question in hours and kilometres is a journey question. Convert to match, once, at the start.
- If the answer comes out over 100 m/s you have converted the wrong way - that is faster than any train.

## Time and work, pipes and cisterns

The single rule for this whole sub-topic: never work with days, always work with ONE DAY'S WORK. If A finishes a job in  $x$  days, A does  $1/x$  of it in a day. Add the one-day works to combine workers, subtract to find one worker from a pair, and turn the total back into days at the very end by taking the reciprocal. Pipes and cisterns is the identical sum, except that an emptying pipe or a leak carries a minus sign.

### Work and pipes - the standard results

| Situation                | Rule                                    | Example                                |
|--------------------------|-----------------------------------------|----------------------------------------|
| A alone takes $x$ days   | one day's work = $1/x$                  | $x = 12$ gives $1/12$                  |
| A and B together         | $1/x + 1/y$ , so time = $xy/(x + y)$    | 12 and 24 give 8 days                  |
| B alone, given A and A+B | $1/y = 1/(A+B \text{ time}) - 1/x$      | 10 and 15 give 30 days                 |
| Men, days, hours, work   | $M_1 D_1 H_1 / W_1 = M_2 D_2 H_2 / W_2$ | 12 men $\times$ 18 days = 216 man-days |
| More men, same job       | days fall in inverse proportion         | 27 men need $216/27 = 8$ days          |

| Situation                | Rule                                              | Example                               |
|--------------------------|---------------------------------------------------|---------------------------------------|
| Filling pipe             | rate = $+1/x$                                     | —                                     |
| Emptying pipe or leak    | rate = $-1/y$                                     | —                                     |
| Filling pipe with a leak | net rate = $1/x - 1/y$                            | 10 and 15 give $1/30$ , i.e. 30 hours |
| Efficiency ratio         | if A is twice as fast as B, A takes half the days | times are inverse to efficiencies     |

Worked example one - A and B together finish a work in 10 days and A alone in 15 days; how long does B alone take? Worked example two - a tap fills a tank in 10 hours but a leak can empty the full tank in 15 hours; with both working, how long does it take to fill? Notice the two sums are numerically identical - that is the point.

- One day's work of A and B together =  $1/10$ . One day's work of A =  $1/15$ .
- One day's work of B =  $1/10 - 1/15 = 3/30 - 2/30 = 1/30$ .
- So B alone takes 30 days. Subtracting the days,  $15 - 10 = 5$ , is the standard error and it is always an option.
- Tank: the tap fills  $1/10$  per hour, the leak empties  $1/15$  per hour.
- Net =  $1/10 - 1/15 = 1/30$  of the tank per hour, so the tank fills in 30 hours.
- Adding instead of subtracting gives 6 hours - impossible, since a leak must make it SLOWER than the tap's own 10 hours.
- Two pipes filling in 6 and 4 hours:  $1/6 + 1/4 = 5/12$  per hour, so  $12/5$  hours = 2 hours 24 minutes = 144 minutes.



**YAAD RAKHO**

Remember - the LCM method kills fractions. Take the total work as the LCM of the given days. A takes 12 days and B 24 days: LCM = 24 units of work, so A does 2 units a day and B does 1. Together 3 units a day, so  $24/3 = 8$  days. No fractions anywhere. Same for tanks: tap 10 h, leak 15 h, LCM 30 litres, tap +3 and leak -2, net +1 litre an hour, so 30 hours.

## Calendars and clocks

Two short topics that together give a guaranteed question, and both run on one idea each. Calendar runs on ODD DAYS - the days left over after taking out complete weeks. Clock runs on the fact that the minute hand gains on the hour hand at 5.5 degrees a minute. Learn the two counting tables below and neither question can take you more than a minute.

### Calendar odd days and clock angles

| Item                   | Value                                                   | Note                                 |
|------------------------|---------------------------------------------------------|--------------------------------------|
| Ordinary year          | 1 odd day                                               | $365 = 52 \text{ weeks} + 1$         |
| Leap year              | 2 odd days                                              | $366 = 52 \text{ weeks} + 2$         |
| 100 years              | 5 odd days                                              | 24 leap + 76 ordinary years          |
| 200 / 300 / 400 years  | 3 / 1 / 0 odd days                                      | the calendar repeats every 400 years |
| Day code from odd days | 0 Sun, 1 Mon, 2 Tue, 3 Wed, 4 Thu, 5 Fri, 6 Sat         | count from Sunday                    |
| Leap year test         | divisible by 4; a century year must be divisible by 400 | 1900 was NOT a leap year, 2000 was   |
| Month codes Jan to Dec | 0, 3, 3, 6, 1, 4, 6, 2, 5, 0, 3, 5                      | add to the year's odd days           |

| Item                    | Value                                                       | Note                           |
|-------------------------|-------------------------------------------------------------|--------------------------------|
| Angle between the hands | $ 30H - 5.5M $ degrees                                      | if over 180, subtract from 360 |
| Hand speeds             | hour 0.5 deg/min, minute 6 deg/min                          | minute hand gains 5.5 deg/min  |
| In 12 hours             | coincide 11 times, opposite 11 times, right angles 22 times | in 24 hours, double each       |

Worked example one - the Constitution came into force on 26 January 1950; what day of the week was it? Worked example two - at what time between 4 and 5 o'clock do the two hands coincide? The first is the full odd-day drill, the second is the gain-of-5.5-degrees drill.

- Break 1950 into 1600 + 300 + 49 completed years, then the days of January.
- 1600 years give 0 odd days; 300 years give 1 odd day.
- The 49 years 1901 to 1949 hold 12 leap years and 37 ordinary years, so  $12 \times 2 + 37 \times 1 = 61$  days = 8 weeks + 5 odd days.
- January up to the 26th is 26 days = 3 weeks + 5 odd days.
- Total odd days =  $0 + 1 + 5 + 5 = 11$ , and  $11 - 7 = 4$ . Code 4 counted from Sunday is Thursday. 26 January 1950 was a Thursday.
- Clock: at 4 o'clock the hour hand stands at 120 degrees and the minute hand at 0, so a gap of 120 degrees must be closed.
- The gap closes at  $6 - 0.5 = 5.5$  degrees a minute, so the time is  $120 \div 5.5 = 240/11 = 21$  and  $9/11$  minutes past 4.

### Finding the day of the week for any date

Split the year as 400s + 100s + the completed years since the last century.

Odd days: 400 years give 0, 300 give 1, 200 give 3, 100 give 5.

For the completed years, count leap years  $L$  and ordinary years  $O$  and take  $2L + O$ .

Add the days elapsed in the current year, month by month, up to the given date.

Add everything and divide by 7; keep the remainder only.

Read the remainder as 0 Sunday, 1 Monday, and so on up to 6 Saturday.

#### TRAP

Three traps in one box. First, 15 August 2026 is a Saturday, so 15 August 2027 is a Sunday - the stretch contains February 2027, an ordinary year, so only ONE odd day is added; students add two out of habit. Second, at 3:30 the angle is 75 degrees, not 90 - by half past, the hour hand has itself crawled half way from 3 towards 4. Third, a century year is a leap year only if it divides by 400, so 1900 had 365 days.

#### REVISION RECAP

1. Square: area  $a^2$ , diagonal  $a\sqrt{2}$ , area from diagonal  $d^2/2$ .  
Rectangle: area  $lb$ , diagonal  $\sqrt{l^2 + b^2}$ .
2. Heron:  $s = (a+b+c)/2$ , area =  $\sqrt{s(s-a)(s-b)(s-c)}$ . The 13-14-15 triangle gives 84.
3. One-half belongs to triangle, trapezium and rhombus. Parallelogram is base  $\times$  height, no half.
4. Circle  $2\pi r$  and  $\pi r^2$ ; sector area =  $(\theta/360)\pi r^2$  and arc =  $(\theta/360)2\pi r$ .
5. Path  $w$  wide outside a rectangle =  $2w(l+b) + 4w^2$ ; inside, subtract the  $4w^2$ .
6. Change every linear dimension by  $k$ : length  $\times k$ , area  $\times k^2$ , volume  $\times k^3$ .  
Side up 20% means area up 44%.

7. Cone: find  $l = \sqrt{r^2 + h^2}$  first. CSA  $\pi r l$ , TSA  $\pi r (l+r)$ , volume one-third  $\pi r^2 h$ .
8. Hemisphere total surface =  $3\pi r^2$ , because the flat face counts. Sphere =  $4\pi r^2$ .
9. Go back to the TOTAL in every average question: total = average  $\times$  number of items.
10. Weighted average =  $(n_1a_1 + n_2a_2)/(n_1+n_2)$ . Never average two group averages of unequal size.
11. Equal distances: average speed =  $2xy/(x+y)$ . Equal times:  $(x+y)/2$ . 60 and 40 give 48, not 50.
12. Grouped mean =  $\Sigma fx / \Sigma f$ . Median needs the cumulative frequency; mode is the largest frequency.
13. Mode =  $3 \text{ Median} - 2 \text{ Mean}$ ; Range = maximum - minimum.
14. Pie chart: 360 degrees = the whole total, so find the value of one degree first. Percentages need only the angles.
15. Percentage change divides by the ORIGINAL value, never the new one.
16. Order matters means  $nPr$ ; order does not means  $nCr$ . Independent choices multiply, they never add.
17. BANANA =  $6!/(3!2!)$  = 60; handshakes among  $n$  =  $nC_2$ ; diagonals of an  $n$ -gon =  $n(n-3)/2$ .
18. Two dice: 36 outcomes, sum ladder 1-2-3-4-5-6-5-4-3-2-1, doublet  $1/6$ , sum 8 is  $5/36$ . Six red face cards.
19. Train past a pole covers its own length; past a platform, length plus platform.  $5/18$  to go from km/h to m/s.
20. Work in one day's work  $1/x$ , never in days. Leak subtracts. Odd days: ordinary 1, leap 2, 400 years 0. Clock angle =  $|30H - 5.5M|$ .